

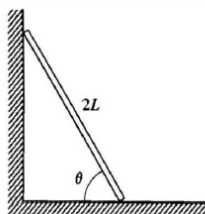
8.012 Problem Set 8

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Problem 4

Kleppner & Kolenkow, Problem 6.41



6.41 This problem utilizes most of the important laws introduced so far and it is worth a substantial effort. However, the problem is tricky (although not really complicated), so don't be alarmed if the solution eludes you.

A plank of length $2L$ leans against a wall. It starts to slip downward without friction. Show that the top of the plank loses contact with the wall when it is at two-thirds of its initial height.

Hint: Only a single variable is needed to describe the system. Note the motion of the center of mass.

Let $\theta(t)$ be the angle the plank makes with the horizontal at time t . This single angle completely specifies the configuration of the system.

The bottom end A slides along the floor and the top end B slides down the wall. The center of mass G of the uniform plank lies at its midpoint.

Kinematics of the centre of mass

The coordinates of G are

$$x_G = L \cos \theta, \quad y_G = L \sin \theta.$$

Differentiating with respect to time,

$$\dot{x}_G = -L \sin \theta \dot{\theta}, \quad \ddot{x}_G = -L(\cos \theta \dot{\theta}^2 + \sin \theta \ddot{\theta}),$$

$$\dot{y}_G = L \cos \theta \dot{\theta}, \quad \ddot{y}_G = -L(\sin \theta \dot{\theta}^2 - \cos \theta \ddot{\theta}).$$

Energy conservation

The plank starts from rest. Taking the gravitational potential energy reference at the floor, the potential energy is

$$U = mgL \sin \theta.$$

At any instant the plank undergoes planar rotation about the instantaneous point of contact with the floor. The moment of inertia of a uniform rod of length $2L$ about an end is

$$I_A = \frac{1}{3}m(2L)^2 = \frac{4}{3}mL^2.$$

Hence the kinetic energy is

$$K = \frac{1}{2}I_A\dot{\theta}^2 = \frac{2}{3}mL^2\dot{\theta}^2.$$

Conservation of mechanical energy gives

$$mgL \sin \theta_0 = mgL \sin \theta + \frac{2}{3}mL^2\dot{\theta}^2,$$

where θ_0 is the initial angle.

Solving for $\dot{\theta}^2$,

$$\dot{\theta}^2 = \frac{3g}{2L}(\sin \theta_0 - \sin \theta). \quad (1)$$

Condition for loss of contact

The only horizontal force on the plank is the normal force from the wall. When the top end loses contact with the wall, this normal force vanishes.

The horizontal equation of motion of the centre of mass is

$$m\ddot{x}_G = -N_w.$$

Thus, at the instant of separation,

$$N_w = 0 \quad \implies \quad \ddot{x}_G = 0.$$

Using the expression for $\ddot{x}_G$,

$$\cos \theta \dot{\theta}^2 + \sin \theta \ddot{\theta} = 0,$$

which gives

$$\ddot{\theta} = -\cot \theta \dot{\theta}^2. \quad (2)$$

Determination of the separation angle

Differentiate the energy equation (1) with respect to time:

$$0 = mgL \cos \theta \dot{\theta} + \frac{4}{3}mL^2\dot{\theta}\ddot{\theta}.$$

Dividing by $m\dot{\theta}$,

$$\ddot{\theta} = -\frac{3g}{4L} \cos \theta. \quad (3)$$

At the instant of separation, both (2) and (3) must hold. Equating them,

$$\cot \theta \dot{\theta}^2 = \frac{3g}{4L} \cos \theta \quad \implies \quad \dot{\theta}^2 = \frac{3g}{4L} \sin \theta. \quad (4)$$

Using (1) and (4),

$$\frac{3g}{2L}(\sin \theta_0 - \sin \theta_c) = \frac{3g}{4L} \sin \theta_c.$$

Solving,

$$2(\sin \theta_0 - \sin \theta_c) = \sin \theta_c \quad \implies \quad \sin \theta_c = \frac{2}{3} \sin \theta_0.$$

Height at separation

The height of the top end is $h = 2L \sin \theta$. Thus,

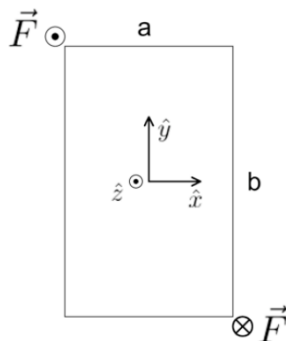
$$\frac{h}{h_0} = \frac{2L \sin \theta_c}{2L \sin \theta_0} = \frac{2}{3}.$$

Final Answer

The plank loses contact with the wall when the top has descended to $\frac{2}{3}$ of its initial height.

Problem 8

8. Rectangular Symmetry. [15 points]



A flat, uniform rectangle with sides of length a and b is at rest in free space (there are initially no external forces acting). You strike two opposite corners with equal and opposite forces $\vec{F}$ as indicated for a short time period Δt .

- (a) [5 pts] Calculate the moments of inertia of the rectangle about horizontal ($\hat{x}$) and vertical ($\hat{y}$) axes passing through its center of mass. Note that this does not require calculating the moment of inertia tensor. Verify your answers by showing that these moments of inertia satisfy the perpendicular axis theorem, $I_x + I_y = I_z$, where I_z is the moment of inertia about an axis coming out of the page ($\hat{z}$) through the center of mass (which you should also calculate).
- (b) [10 pts] Calculate the spin vector ($\vec{\omega}$) and angular momentum vector ($\vec{L}$) immediately after the strike. Show that these are not parallel, and that the spin vector points along the opposite diagonal from where the forces were applied if $a \neq b$.

Let Mass of the Rectangle be m

a.

By definition of Moment of Inertia,

$$I_x = \int_{-\frac{b}{2}}^{\frac{b}{2}} y^2 \frac{m}{ab} a dy = \frac{m}{b} \frac{b^3}{12} = \frac{mb^2}{12}$$

Similarly,

$$I_y = \frac{ma^2}{12}$$

By definition of Moment of Inertia,

$$\begin{aligned} I_z &= \int_{y=-\frac{b}{2}}^{\frac{b}{2}} \int_{x=-\frac{a}{2}}^{\frac{a}{2}} (\sqrt{x^2 + y^2})^2 \frac{m}{ab} dx dy \\ &= \int_{y=-\frac{b}{2}}^{\frac{b}{2}} \frac{m}{ab} \left(\frac{a^3}{12} + y^2 a \right) dy = \int_{y=-\frac{b}{2}}^{\frac{b}{2}} \frac{m}{b} \left(\frac{a^2}{12} + y^2 \right) dy \\ &= \frac{m}{b} \left(\frac{a^2 b}{12} + \frac{b^3}{12} \right) = m \left(\frac{a^2}{12} + \frac{b^2}{12} \right) \end{aligned}$$

Therefore,

$$\boxed{I_x = \frac{mb^2}{12} \quad I_y = \frac{ma^2}{12} \quad I_z = \frac{m(a^2 + b^2)}{12} = I_x + I_y}$$

b.

Let the Forces be $F\hat{z}$ and $-F\hat{z}$

So the distance between the application points is the Vector $\vec{d} = -a\hat{x} + b\hat{y}$

Therefore, Moment of the Force is

$$\begin{aligned} \tau &= \vec{d} \times \vec{F} \\ &= (-a\hat{x} + b\hat{y}) \times (F\hat{z}) = F(b\hat{x} + a\hat{y}) \end{aligned}$$

Hence,

$$\boxed{L = \tau(\Delta t) = F(\Delta t)(b\hat{x} + a\hat{y})}$$

Note that the Rectangle is Symmetric about x, y, z axes. So the Inertia tensor is given by

$$I = \begin{bmatrix} I_x & 0 & 0 \\ 0 & I_y & 0 \\ 0 & 0 & I_z \end{bmatrix} = \begin{bmatrix} \frac{mb^2}{12} & 0 & 0 \\ 0 & \frac{ma^2}{12} & 0 \\ 0 & 0 & \frac{m(a^2 + b^2)}{12} \end{bmatrix} = \frac{m}{12} \begin{bmatrix} b^2 & 0 & 0 \\ 0 & a^2 & 0 \\ 0 & 0 & a^2 + b^2 \end{bmatrix}$$

We know that,

$$L = I\omega$$

Therefore,

$$\omega = I^{-1}L = \frac{12}{m} \begin{bmatrix} \frac{1}{b^2} & 0 & 0 \\ 0 & \frac{1}{a^2} & 0 \\ 0 & 0 & \frac{1}{a^2+b^2} \end{bmatrix} F(\Delta t) \begin{bmatrix} b \\ a \\ 0 \end{bmatrix} = \frac{12F(\Delta t)}{m} \begin{bmatrix} \frac{1}{b} \\ \frac{1}{a} \\ 0 \end{bmatrix}$$

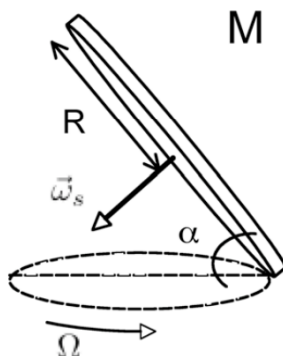
Hence,

$$\omega = \frac{12F(\Delta t)}{m} \left(\frac{\hat{x}}{b} + \frac{\hat{y}}{a} \right)$$

ω is Not Parallel to L if $a \neq b$

Problem 9

9. Euler's Disk. [15 points]



If you spin a coin (a solid uniform disk) about a vertical axis on a table, it will eventually lose energy and begin to wobble. Wobbling motion can basically be described as the rolling of the coin without slipping while tilted from horizontal by an angle α . Assume that once the coin starts to wobble, it falls sufficiently slowly tht you can assume α is constant and the center of mass remains at a fixed point. Let ω_s be the spin frequency of the coin, and Ω be the frequency at which the contact point traces a circle on the table.

- (a) [5 pts] Show that the total angular velocity vector $\vec{\omega}$ of the coin has magnitude $\Omega \sin \alpha$ and points in a direction from the contact point up through the center of mass of the coin.

- (b) [5 pts] Show that:

$$\Omega = 2\sqrt{\frac{g}{L \sin \alpha}} \quad (1)$$

and discuss what happens at the limits of α . It helps to verify that the direction of $\vec{\omega}$ is along one of the principal axes of the coin.

- (c) [5 pts] Show that, when viewed from above, the face of the coin appears to rotate with frequency

$$\Omega = 2(1 - \cos \alpha)\sqrt{\frac{g}{L \sin \alpha}}. \quad (2)$$

Verify these results by spinning a coin yourself!

a.

By condition of pure rolling,

$$v_{\text{CM}} + \omega \times r_{\text{Contact/CM}} = 0$$

Since, C.O.M. remains at a fixed point,

$$v_{\text{CM}} = 0 \implies \omega \times r_{\text{Contact/CM}} = 0$$

$$\implies \omega \parallel r_{\text{Contact/CM}}$$

By Geometry,

$$\Omega R \sin(\alpha) = \omega R \implies \omega = \Omega \sin(\alpha)$$

Hence,

Magnitude of ω is $\Omega \sin(\alpha)$

Direction is from the Contact Point up through the C.O.M.

b.

Here, ω is along a Principle axis of the coin. So

$$L \parallel \omega$$

ω is along the Diameter. So

$$I = \frac{MR^2}{4} \implies L = I\omega = \frac{MR^2\Omega \sin(\alpha)}{4}$$

The Magnitude of L remains constant. Only the Direction changes. Therefore there is an existence of Torque τ Perpendicular to L . The rate of change of L is given by

$$\frac{dL}{dt} = \Omega \times L = \frac{MR^2\Omega^2 \sin(\alpha) \cos(\alpha)}{4}$$

Forces acting on the coin are:- 1. Gravity 2. Contact Forces

τ about Contact Point is caused only due to Gravity.

Therefore,

$$\tau = |R(-\cos(\alpha)\hat{y} + \sin(\alpha)\hat{z}) \times -Mg\hat{z}| = MgR \cos(\alpha)$$

So

$$\frac{MR^2\Omega^2 \sin(\alpha) \cos(\alpha)}{4} = MgR \cos(\alpha)$$

Hence

$$\Omega = 2\sqrt{\frac{g}{R \sin(\alpha)}}$$

Limit $\alpha \rightarrow 0$.

As the disk becomes nearly horizontal,

$$\sin \alpha \rightarrow 0 \implies \Omega \rightarrow \infty.$$

Thus, in the idealized model the angular speed diverges as the disk flattens. In practice, dissipative effects such as friction and air resistance prevent this divergence.

Limit $\alpha \rightarrow \frac{\pi}{2}$.

As the disk becomes nearly vertical,

$$\sin \alpha \rightarrow 1 \implies \Omega \rightarrow 2\sqrt{\frac{g}{R}}.$$

In this limit the angular speed approaches a finite constant, corresponding to ordinary rolling motion.

c.

We seek the apparent rotation rate of the coin's face as viewed from above, i.e. the rate at which a marked diameter on the face rotates about the vertical axis.

The motion consists of two simultaneous effects:

- Precession of the disk's symmetry axis about the vertical with angular speed Ω .
- Rotation of the disk relative to its own plane due to rolling without slipping.

The precession alone would cause the face to rotate forward at rate Ω .

From the rolling without slipping constraint ($\vec{v}_{\text{contact}} = 0$), standard kinematics for a tilted rolling disk gives the relative spin rate of the disk about its symmetry axis as

$$\omega = \Omega \cos \alpha,$$

with the sense such that this rotation opposes the apparent rotation produced by precession.

Therefore, the net apparent angular speed of the face is

$$\Omega_{\text{face}} = \Omega - \omega = \Omega(1 - \cos \alpha).$$

Substituting the result from part (b),

$$\Omega = 2\sqrt{\frac{g}{R \sin \alpha}},$$

we obtain

$$\Omega_{\text{face}} = 2(1 - \cos \alpha)\sqrt{\frac{g}{R \sin \alpha}}$$